Signed graph forest closure

Ten main declarations and nine new controls passed fresh replay, with nineteen exact standard-axiom readbacks. Eight modules returned native/gate exit zero. A separate false equality returned native/gate exit one.

Constructed forest witness. Actual parent reachability identifies the child component. Add rank(parent)+1 to precisely that component. Every unchanged edge retains its decrease; the new edge strictly decreases to the surviving root. The proof derives forest preservation, child attachment and transfer of all old child-component paths to the surviving root.

Native edge composition. The first returned root survives the second find. The composed theorem derives native reduced coefficients from the original edge and preserves Forest and all vertex/anchor equations for either union orientation. A later find has a finite successful execution. Assumptions include a valid initial forest/state, a true edge, distinct returned roots and unit-square decoded signs.

Controls. Explicit signed-descendant finds feed both orientations. Root-to-descendant attachment forms a cycle under every rank. A nonroot child can safely be cut and attached to a distinct root, and this stronger accepted case is distinguished from unsafe nonroot-parent attachment.

Remaining interfaces. Native size metadata, allocation, word/sign and array/counter bounds, compiled equivalence and the full same-root edge branch remain separate. The rank is a mathematical witness, not a native size update or complexity bound.

Custody. Source-first plan, exact native binding, fresh invocation directories and retained failures are included. Independent review follows publication. Cached toolchain dependencies are trusted. No scientific execution or repository change occurred.